

1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$g(a+h) \approx g(a) + g'(a) \cdot h.$$

$$\text{Substitute } a=2, h=0.1, g(a)=5, g'(a)=-2$$

$$\Rightarrow g(2.1) \approx 5 + (-2) \cdot 0.1 = 4.8$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

$$\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2} = \lim_{x \rightarrow 2} \frac{g(x) - g(2)}{x - 2} = g'(2) = -2.$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$\begin{aligned} f'(x) &= (x^2)' \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \left(\sin\left(\frac{1}{x}\right)\right)' \\ &= 2x \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) \\ &= 2x \cdot \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right). \end{aligned}$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$\begin{aligned} f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{x^2 \cdot \sin\left(\frac{1}{x}\right) - 0}{x - 0} \\ &= \lim_{x \rightarrow 0} x \cdot \sin\left(\frac{1}{x}\right). \end{aligned}$$

Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

$$\Rightarrow \underline{-|x|} \leq x \cdot \sin\left(\frac{1}{x}\right) \leq \underline{|x|} \quad \star$$

Since $\lim_{x \rightarrow 0} |x| = 0 = \lim_{x \rightarrow 0} -|x|$

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$$\lim_{x \rightarrow 0} x \cdot \sin\left(\frac{1}{x}\right) = 0.$$

So $f'(0) = 0.$

3. (6 points) An object moves along a line with position $s(t) = t e^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$\begin{aligned} s'(t) &= 1 \cdot e^{kt} + t \cdot e^{kt} \cdot k \\ &= (kt+1) e^{kt} \end{aligned}$$

$$\begin{aligned} s''(t) &= k \cdot e^{kt} + (kt+1) \cdot e^{kt} \cdot k \\ &= (k^2 t + 2k) e^{kt}. \end{aligned}$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$s''(0) = (k^2 \cdot 0 + 2k) \cdot e^{k \cdot 0} = 2k.$$

$$\text{Since } s''(0) = 4.$$

$$2k = 4$$

$$k = 2$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

Take derivative. $\frac{2x}{4} + \frac{2y \cdot y'}{8} = 0$.

$$\frac{x}{2} + \frac{yy'}{4} = 0$$

$$\frac{yy'}{4} = -\frac{x}{2}$$

$$y' = -\frac{x}{2} \cdot \frac{4}{y} = -\frac{2x}{y}$$

At $(-\sqrt{2}, 2)$. $y' = -\frac{2x}{y} = -\frac{2 \cdot (-\sqrt{2})}{2} = \sqrt{2}$

Tangent line $y - 2 = \sqrt{2}(x + \sqrt{2})$.

$$\Rightarrow y = \sqrt{2}x + 4$$